

PRACTICAL 2 DAA

#LINEAR SEARCH

```
import time

# Take list input from user
numbers = list(map(int, input("Enter the numbers (space separated): ").split()))

# Take element to search
key = int(input("Enter the number to search: "))

# Start execution time
start = time.time()

found = False

# Linear Search
for i in range(len(numbers)):
    if numbers[i] == key:
        print("Element found at index:", i)
        found = True
        break

if not found:
    print("Element not found")

# End execution time
end = time.time()

# Execution time
print("Execution Time:", end - start, "seconds")

# Time and Space Complexity
print("Time Complexity: O(n)")
print("Space Complexity: O(1)")
```

output

```
Enter the numbers (space separated): 34 54 67 76 8
Enter the number to search: 8
Element found at index: 4
Execution Time: 0.0003676414489746094 seconds
Time Complexity: O(n)
Space Complexity: O(1)
```

#BINARY SEARCH

```
import time

# Take sorted list input from user
numbers = list(map(int, input("Enter sorted numbers (space separated): ").split()))
```

```

# Take element to search
key = int(input("Enter the number to search: "))

# Start execution time
start = time.time()

low = 0
high = len(numbers) - 1
found = False

while low <= high:
    mid = (low + high) // 2

    if numbers[mid] == key:
        print("Element found at index:", mid)
        found = True
        break
    elif numbers[mid] < key:
        low = mid + 1
    else:
        high = mid - 1

if not found:
    print("Element not found")

# End execution time
end = time.time()

# Execution time
print("Execution Time:", end - start, "seconds")

# Time and Space Complexity
print("Time Complexity: O(log n)")
print("Space Complexity: O(1)")

```

OUTPUT

```

Enter sorted numbers (space separated): 23 45 76 56 98
Enter the number to search: 23
Element found at index: 0
Execution Time: 0.0004949569702148438 seconds
Time Complexity: O(log n)
Space Complexity: O(1)

```